

Exploratory Data Analysis and Machine Learning on Movie Titles Dataset

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Abstract—This paper presents an analysis of a movie and TV show dataset using Python. The study focuses on data cleaning, exploratory data analysis (EDA), and visualization to uncover patterns in content types, genres, and ratings. Additionally, a machine learning model is implemented to predict IMDb scores based on various features. The results highlight trends in content production and provide insights into audience preferences.

INTRODUCTION

With the rapid growth of digital streaming platforms, large amounts of content data are generated. Analyzing such data helps in understanding trends, user preferences, and industry patterns. In this project, a dataset containing information about movies and TV shows is analyzed using Python. The objective is to explore patterns in content production, genres, and ratings, and to build a predictive model for IMDb scores.

EASE OF USE

The implementation of this project is simple and user-friendly due to the use of Python and its powerful libraries. Python provides an easy-to-understand syntax, making it suitable for beginners and students.

Libraries such as Pandas and NumPy simplify data manipulation and preprocessing tasks. Data visualization is effectively handled using Matplotlib and Seaborn, which allow clear representation of patterns and trends.

Furthermore, Scikit-learn provides a straightforward interface for implementing machine learning models. The Linear Regression model used in this project can be easily trained and evaluated with minimal code.

DATASET DESCRIPTION

The dataset used in this project contains information about movies and TV shows. It includes attributes such as content type, release year, runtime, genres, production countries, and ratings from IMDb and TMDB.

For analysis, only relevant columns were selected, including type, release year, runtime, genres, production countries, IMDb score, TMDB score, and popularity.

The dataset was preprocessed to handle missing values and remove inconsistencies, ensuring accurate and reliable results.

METHODOLOGY

A. Data Cleaning

- Removed missing values in important columns such as IMDb score and release year
- Filled missing values in runtime, genres, and production countries
- Removed duplicate records
- Selected relevant columns for analysis

B. Exploratory Data Analysis (EDA)

- Comparison between movies and TV shows
- Year-wise trend of content production
- Identification of top genres
- Analysis of production countries

C. Data Visualization

Different visualization techniques were used:

- Bar charts for categorical comparison
- Line plots for trend analysis
- Histograms for distribution of ratings

These visualizations helped in identifying patterns clearly.

D. Machine Learning Model

A Linear Regression model was implemented to predict IMDb scores using the following features:

- Release Year
- Runtime
- TMDB Popularity
- TMDB Score

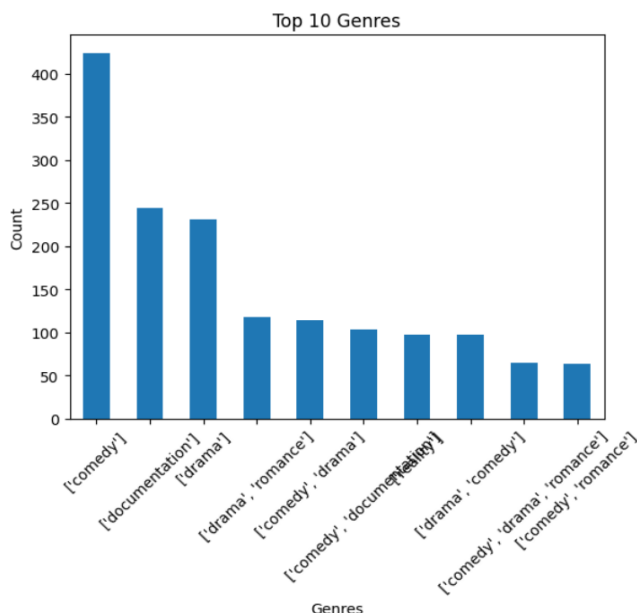
The dataset was divided into training and testing sets. Model performance was evaluated using Mean Squared Error (MSE) and R^2 Score.

RESULTS AND DISCUSSION

The analysis revealed several important insights:

- Movies are more common than TV shows in the dataset
- Content production has increased significantly after the year 2000
- Genres such as Drama and Comedy are the most frequent
- Most IMDb scores are concentrated between 6 and 8
- Popularity distribution indicates that only a few titles are highly popular

The machine learning model showed moderate performance in predicting IMDb scores, indicating a relationship between popularity and ratings.



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CONCLUSION

This project successfully demonstrated the use of Python for data analysis and machine learning. The study identified key trends in the movie dataset and provided meaningful insights into content production and audience preferences.

The implementation of a machine learning model further enhanced the analysis by predicting IMDb scores. Future work can involve advanced models and deeper analysis of textual data

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